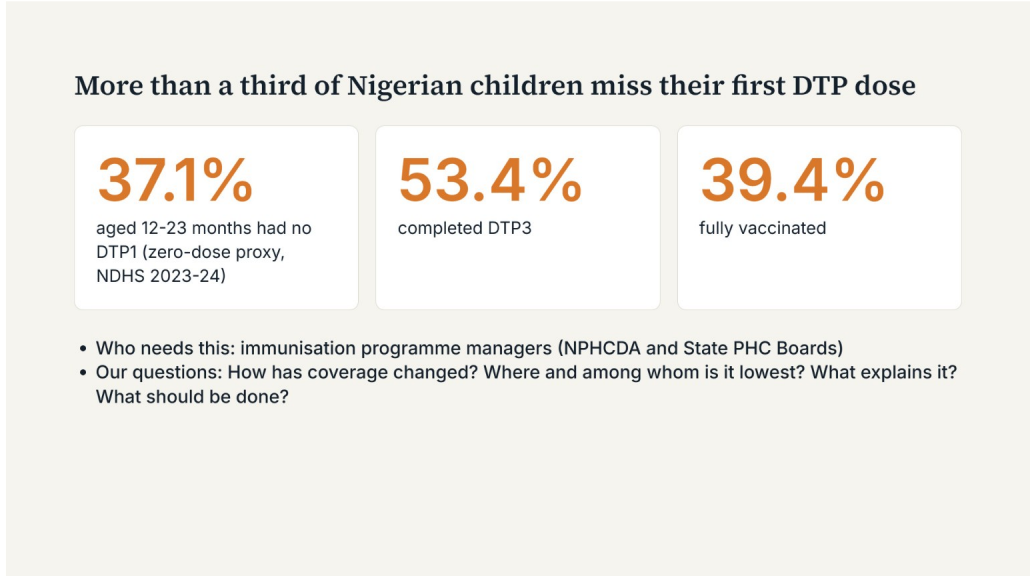
Public Health Analytics | eHA Academy Data Science Capstone

Childhood Immunisation Coverage and Inequalities in Nigeria

A data-driven public health analysis

Team: Moses O. Falodun, Muflihat Abdulmajid, Aisha Hilal Galadanchi | 30 October 2026



Programme managers need to know where and among whom to focus outreach.

Two public sources, cleaned and cross-checked

- WHO/UNICEF WUENIC survey data: 22 Nigerian surveys (1,140 rows)
- DHS Program API: 37 states x 3 surveys = 111 state-survey rows
- 12 data issues documented and fixed (for example 122 blank survey names, zone rows removed)
- The API and WUENIC national values match exactly (difference 0.0)
- Surveys are named by fieldwork year: NDHS 2018 and NDHS 2023-24

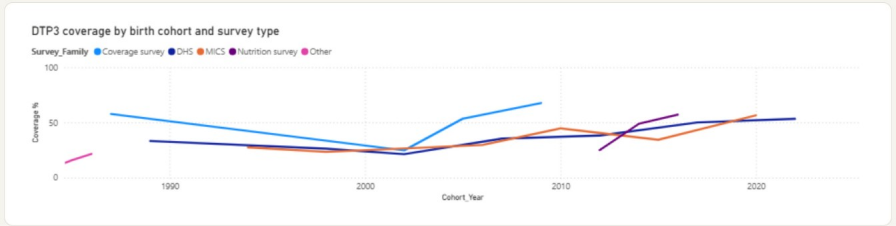
We checked that two independent downloads agree before using either.

Coverage improved modestly, but the zero-dose gap did not close

- DTP3: 53.4% (was 50.1%)
- DTP dropout: 15.1% (was 23.3%), real progress
- Fully vaccinated: 39.4% (was 31.3%)
- Zero-dose proxy (100 - DTP1): 37.1% (was 34.7%), no clear improvement
- Children with no vaccine of any kind: 19.2% to 30.9%
- Surveys disagree: coverage surveys read about 23 points above MICS, so we compare only DHS and MICS
- Context: ranks 28th of 32 African countries with a recent DHS or MICS survey

Children are finishing the series more often, but too many never start it.

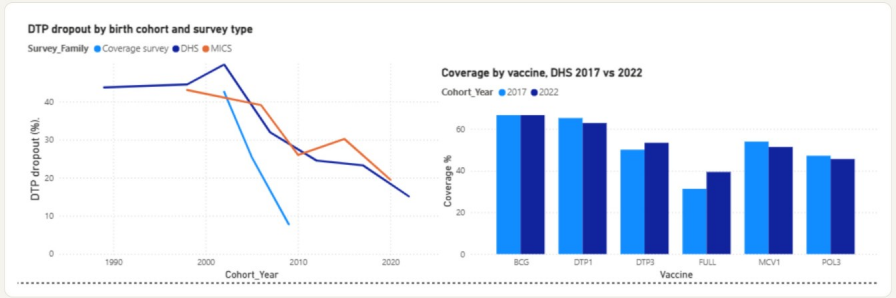
DTP3 coverage by birth cohort and survey type



Source: Power BI dashboard, page 1. Survey types are reported separately and never combined.

Each line is a different survey family, so compare trends within a family rather than across them.

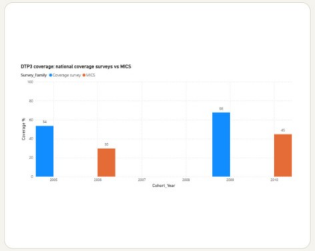
DTP dropout has fallen, and coverage differs by vaccine



Source: Power BI dashboard, page 2.

Children are finishing the series more often. Dropout is the gap between the first and third DTP doses.

Coverage surveys read about 23 points above MICS

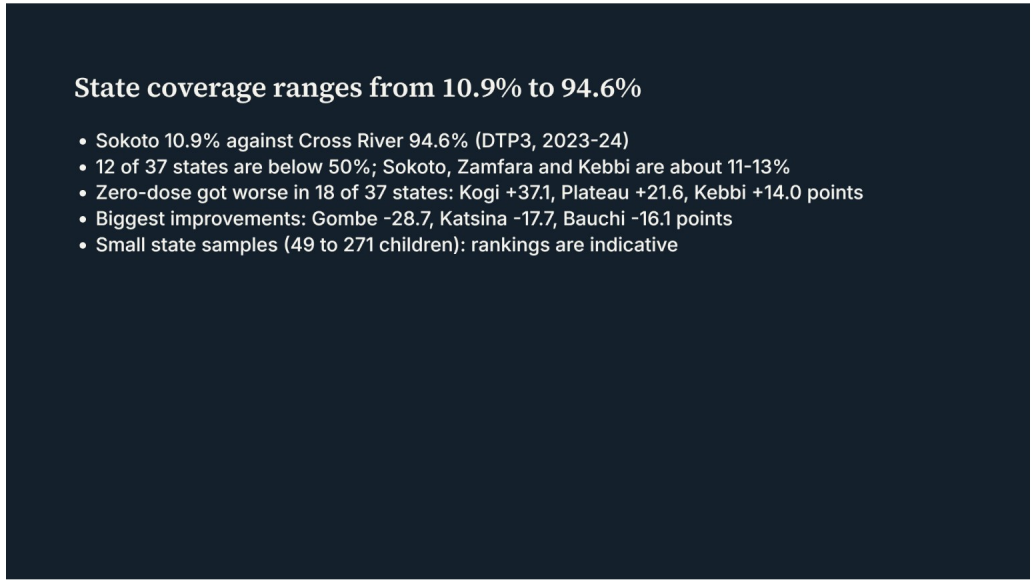


53.5% vs 29.6%

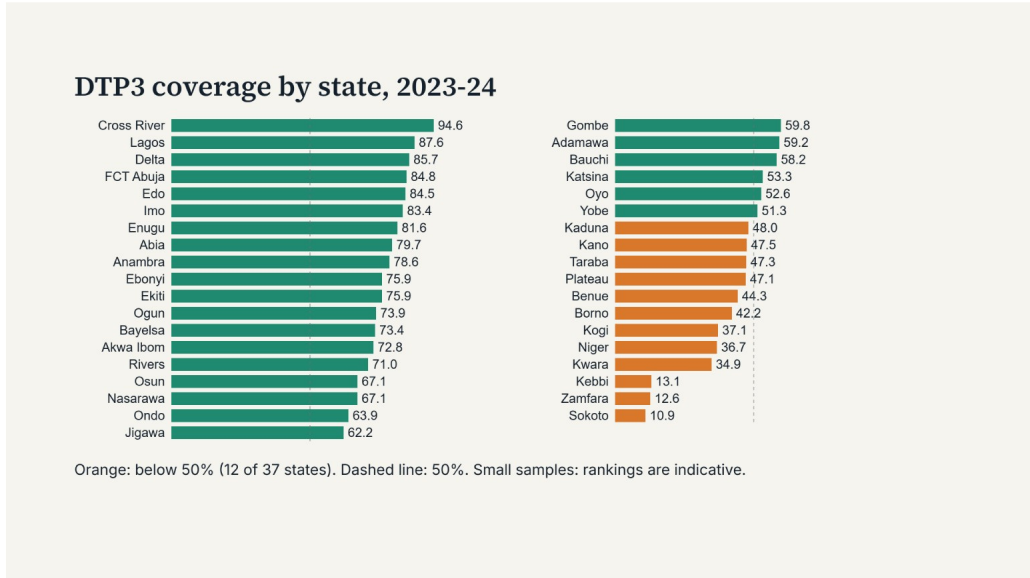
67.7% vs 44.7%

The two survey types measure coverage differently, so we report them separately and compare only DHS and MICS.

Source: Power BI dashboard, page 3.



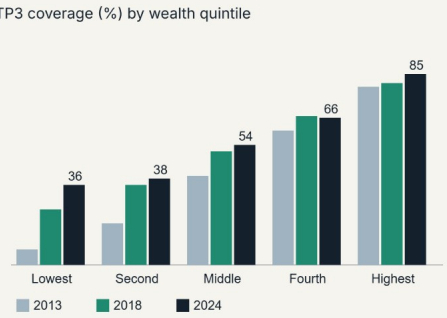
The national average hides a gap of more than 80 points between states.



Twelve states are below 50%. The three lowest are Sokoto, Zamfara and Kebbi.

Poverty and mothers' education drive the gap

- Poorest fifth 35.7% against richest fifth 85.3% (DTP3, 2024)
- Wealth gap narrowed: 72.5, then 56.4, then 49.6 points (2013, 2018, 2024)
- Urban-rural gap narrowed from 37.3 to 19.9 points, partly because urban coverage fell (67.9% to 65.4%)
- Mothers with no education 33.7% against higher education 88.6%



Gaps are narrowing, but the poorest children are still far behind. Demo cue: a good place for the live Power BI demo (pages 1 to 4).

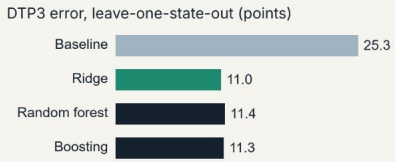
Can simple indicators explain state coverage, and is a complex model needed?

- Data: 111 state-survey rows; targets: DTP3 and zero-dose proxy
- 8 predictors: antenatal care, facility births, skilled birth attendance, women's education, fertility, poorest-quintile share (none from vaccination data)
- Models: training mean (baseline), ridge regression, random forest, gradient boosting
- Validation: leave-one-state-out, and train on 2013 and 2018 to predict 2024
- Controls: leakage check, fixed seed, every run logged

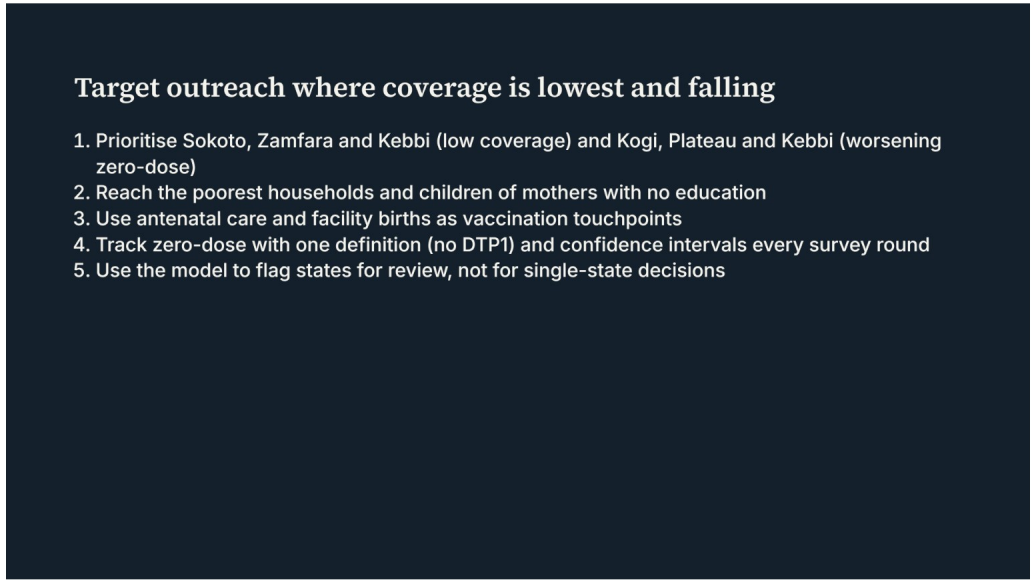
Each test hides whole states or a whole survey year from the model.

A simple model matches complex ones; women's education matters most

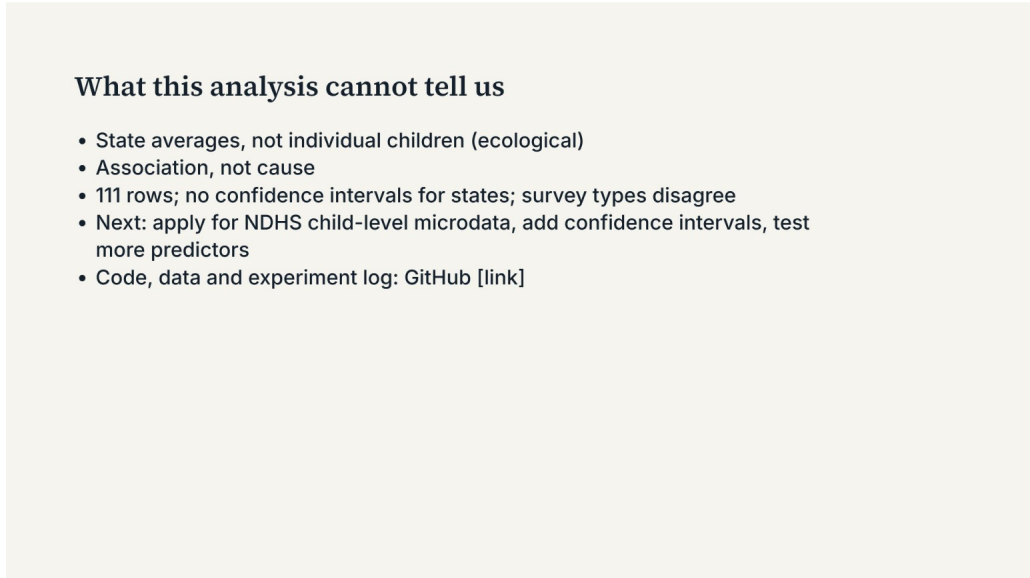
- Unseen-state DTP3 error: 25.3 points (baseline) to 11.0 points (ridge), 56% lower
- Ridge was best in 3 of 4 comparisons; random forest and boosting added nothing reliable
- Women's secondary education is the strongest predictor
- Poorly predicted: Ebonyi, Niger, Bauchi (the model should not be trusted there)



With 111 rows, the simple model is the honest choice.



These show where to look first; they are not proof of cause.



We state the limits plainly so the findings can be used safely.